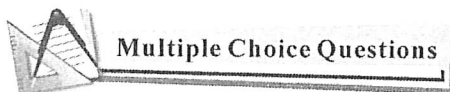


第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)



$$\frac{1}{3} \quad 1, 2$$

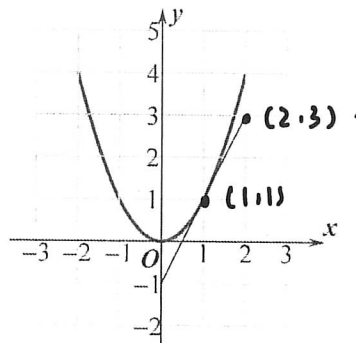
- C 1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

(A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3

- C 2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

(A) 4 (B) -4 (C) -3 (D) 4

- D 3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) = (\quad)$.



(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2 $\frac{k(2^3+4) - (-k+2)}{1} = 3$

- C 4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

(A) $\frac{5}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$ $-8k + 4 + k - 2 = 3$

- C 5. **Calculator** Let $y = \tan^{-1} x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5} =$

(A) -12.056 (B) 0 (C) 271.066 (D) 273.066 $-7k + 2 = 3$
 $-7k = 1$
 $k = -\frac{1}{7}$

- C 6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point? $\sqrt{x^2 + 1} = 3x$ $\because 0 < x < 1$

(A) 0.354 (B) -0.001 (C) -2.666 (D) 0 $x^2 + 1 = 9x^2$ $\therefore x = \frac{\sqrt{5}}{5}$

$$5x^2 = 1$$

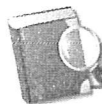
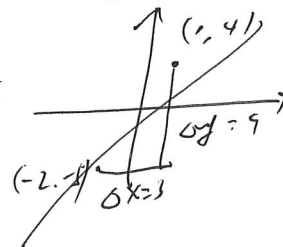
$$x^2 = \frac{1}{5}$$

$$x = \pm \frac{1}{\sqrt{5}} = \pm \frac{\sqrt{5}}{5}$$

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?
- (A) At time $t = 10$, the population of insects is 230.
- (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
- (C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.
- (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.
8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

(A) 3

(B) -3

(C) $\frac{1}{3}$ (D) $-\frac{1}{3}$ 

Free Response Questions

9. Let $f(x) = x^2 - 2x$.(a) Find the average rate of change of f over $[1, 3]$.(b) Find $\frac{f(2+h) - f(2)}{h}$.(c) Find the instantaneous rate of change of f at $x = 2$.

$$a: \frac{f(3) - f(1)}{3 - 1} = \frac{-1 - 1}{-2} = 2$$

$$b: \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2(2+h) - 4 + 4}{h} = \lim_{h \rightarrow 0} \frac{4 + h^2 + 4h - 4 - 2h}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0} h + 2 = 2$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 \\ &= 3x^2 \end{aligned}$$